UEC635: BLOCKCHAIN TECHNOLOGY

L	T	P	Cr
2	0	2	3.0

Course Objective:
This course covers the conceptual and application aspects of fast growing and latest technology Blockchain. The popularity of digital cryptocurrencies has led the foundation of Blockchain, it is a public digital ledger which share the information in a secure way. The various applications of Blockchain are business process management, smart contracts, IoT and so on. In this course fundamental design and architectural primitives of Blockchain, the system and the security aspects will be covered.

Introduction to Blockchain: Blockchain Theory, Immutable Ledger, Smart Networks, Cryptographic Wallets, Blockchain Global Peer to peer software network, Cryptocurrency, Bitcoin mining, Bitcoin scalability, Blockchain risks: Technical, Regulatory, Perception, Payment networks, Blockchain applications, Decoupling Decision-making and Automated execution, Smart Contracts: Bitcoin, Ethereum.

Hyperledger Fabric: Transaction flow, details, membership, identity management, hyperledger composer, application development and network administration,

Blockchain use cases: Blockchain Consensus Algorithms, Byzyantine Fault Tolerance, Applications in finance, supply chain, other industries and Government.,

Miscellaneous: Alt Coins, Ripple, Neo, Litecoin, Cardano, Stellar, Blockchain security and research aspects.

Laboratory Work: Pre-requisite for basics of Blockchain**,** Create a blockchain using Python, Create a cryptocurrency using Python, Create a Smart Contract using Python, Implementation and testing of hyperledger, execution and understanding of bitcoin, implementation, concepts and exposure to blockchain using any language. Modelling, designing and testing of application specific project and research papers.

Familiarization with standards: IEEE 3205-2023 (interoperability/data authentication), IEEE 2418.2-2020 (data formats), IEEE 3201-2024 (access control), and IEEE 3209-2023 (identity key management).

Course Learning Outcomes (CLO):
The student will be able to:
1. Identify the basics of blockchain and its architectures.
2. Know the use of digital currency and consensus algorithms.
3. Implement hyperledger for various real world problems
4. Apply Blockchain technologies in societal, environmental, real world problems.

Text Books:

1. Arvind N., Joseph B., Edward F., Andrew M., and Steven G., "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press, ISBN-13: 978-0691171692.
2. Henning D., and Create S., "Ethereum: Blockchains, Digital Assets, Smart Contracts, Decentralized Autonomous Organizations" Independent Publishing Platform, ISBN-13: 978-1523930470.

Reference Books:

1. Arshdeep B., and Vijay M. "Blockchain Applications: A Hands-on Approach", Vpt, ISBN-13: 978-0996025560.
2. Roger W. "The Science of the Blockchain", Create Space Independent Publishing Platform, 2016, ISBN-978-1522751830

UEC 732: NATURAL LANGUAGE PROCESSING AND APPLICATIONS

L	T	P	Cr
2	0	2	3.0

Course Objective: To familiarize the students with natural language processing (NLP) and introduce major algorithms pertaining to real world problems. Students will be able to design and implement machine learning solutions to NLP problems; and be able to evaluate and interpret the results of the algorithms.

Introduction: Historical overview of Natural Language Processing, Ambiguity and uncertainty in language, Phases in natural language, Word Level analysis: Regular languages and their limitations, Finite-state automata, Morphological parsing, Types of Morphemes, Tokenization, N-grams, Stemming, Lemmatization, Spell checking, Optimal alignment of sequences, String edit operations, Edit distance, and examples of use in spelling correction, and machine translation,

Text Categorization using Classical ML: Classification Methods: Hand-coded rules, Bayesian Methods, Feature-Based Linear Classifiers, Exponential Models, Diving Deeper into Feature Engineering, Evaluation in Multi-class Problems, Decision tree based algorithms in NLP.

Language modeling and Naive Bayes: Markov models, N-grams. Estimating the probability of a word, and smoothing. Generative models of language and their application to building an automatically-trained email spam filter, and automatically determining the language (English, Hindi etc.).

Deep Learning for NLP: Auto encoders, RNNs for Variable Length Sequences, Tricks for Training RNNs, Recurrent neural networks for parts-of-speech tagging, LSTM and GRU networks for NLP applications: Sentiment analysis, Context analysis, and Summarization. Implementation of word2vec: CBOW and Skip-gram, Negative sampling optimization in word2vec, GloVe using gradient descent and alternating least squares, BERT, Tokenization in Transformer Model and Attention.

Familiarization with standards: IEEE 3168-2024 for testing ML-based NLP service robustness against corruption and attacks, P3394 for LLM application interfaces, and IEEE 2841-2022 for deep learning evaluation frameworks.

Laboratory Work:

1. Introduction to MLP Frameworks: NLTK, iNLTK, and Tensorflow (Keras).
2. Tokenization, Segmentation, Parts-of-Speech Tagging and Normalization
3. Language Modelling
4. Text Classification using n-gram models and Sentiment Analysis
5. Obtain pre trained word vectors from GloVe and word2vec and Text classification
6. Parts-of-Speech Tagging Recurrent Neural Network in Tensorflow.
7. Named Entity Recognition RNN in Tensorflow.
8. Implement LSTM, GRU and its variants for NLP applications
9. Implement Tokenization and Attention mechanism in Transformer Models
10. Fine-Tuning BERT for Text Classification

Course Learning Outcomes (CLOs): At the end of the course the student will be able to

1. Explain the linguistic foundations of Natural Language Processing and its computational underpinnings.
2. Apply classical NLP techniques and probabilistic models for a given sequence of words
3. Analyze feature-based and probabilistic text classification methods and evaluation strategies for multi-class NLP problems.
4. Analyze and evaluate deep networks and transformer-based models for semantics, context, and sentiment level language understanding.